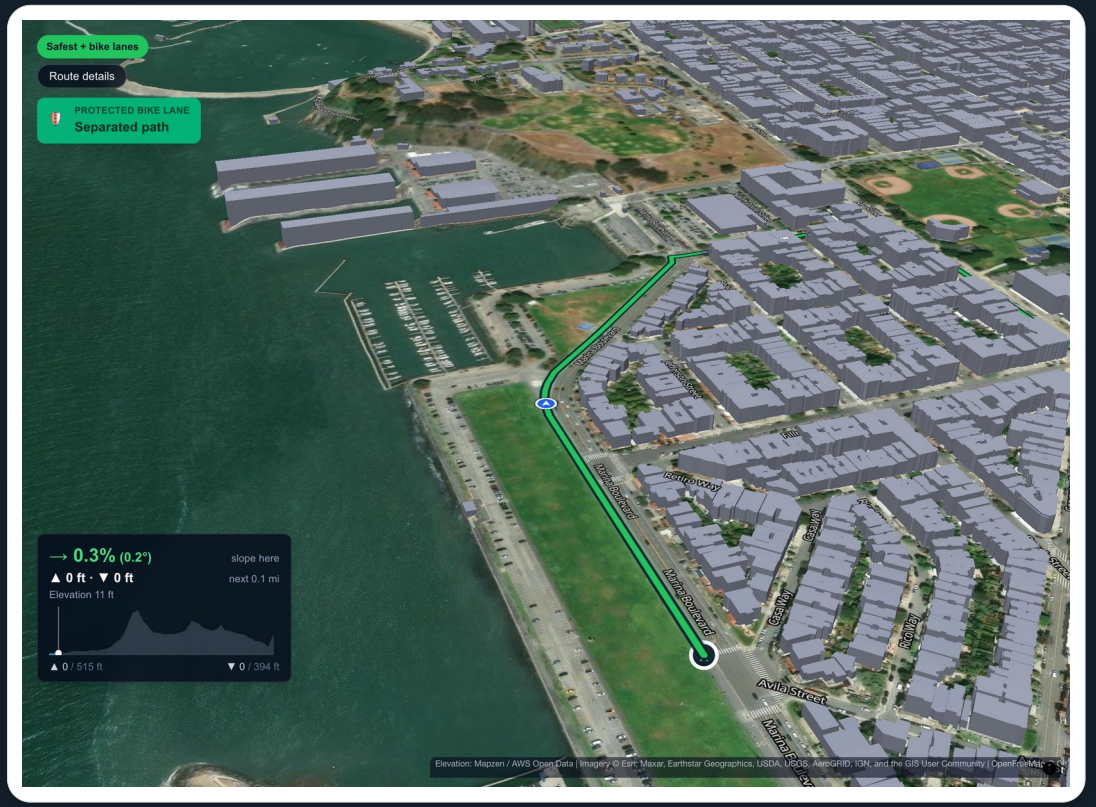
BetterRoutes

The bike map that optimizes routes in San Francisco for safety and custom preferences over speed



Bike danger is concentrated

1,103

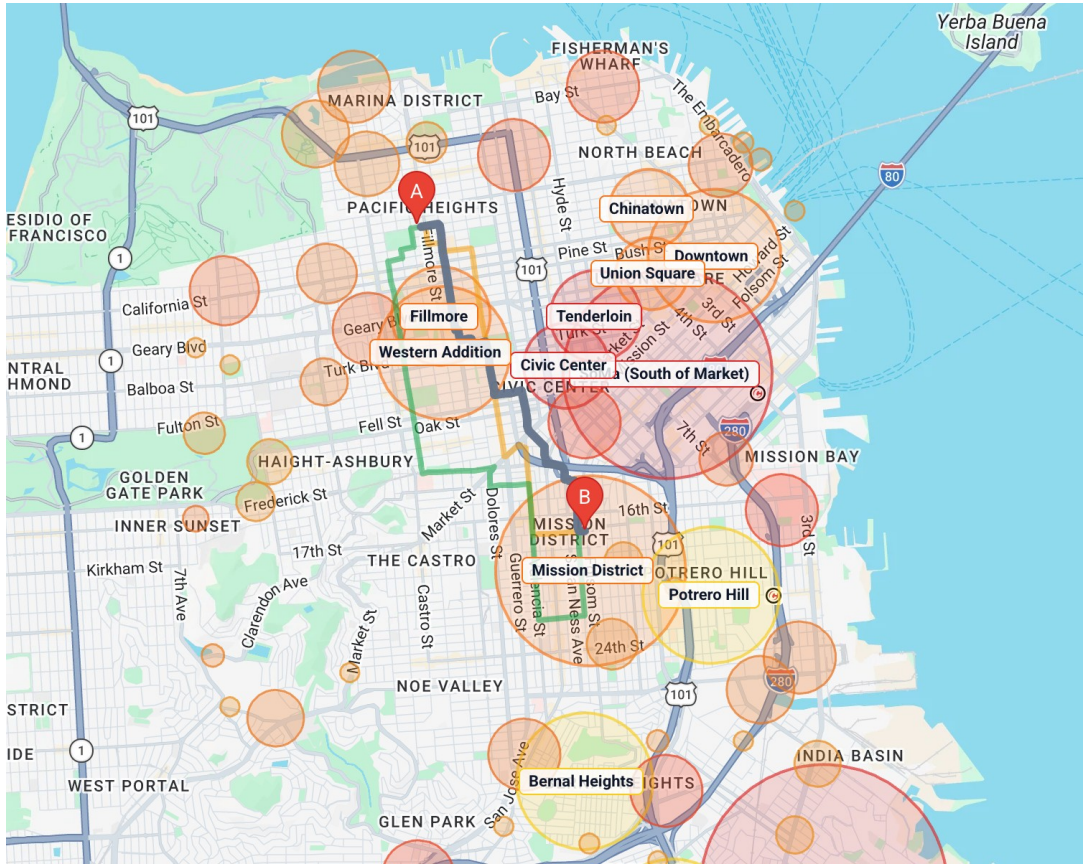
cyclists killed on US roads in 2024

52,887

cyclists injured in 2024 — up 6% from 2023

4% → 55%

4% of SF street miles hold 55% of severe and fatal bike injuries

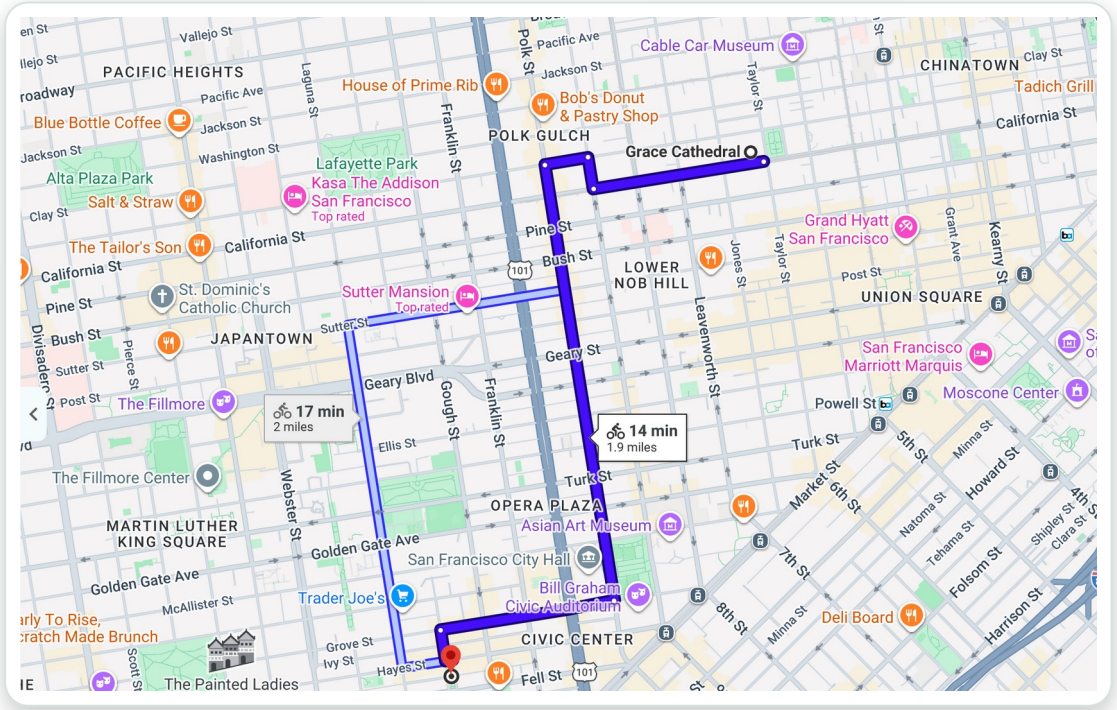


Sources: NHTSA Traffic Safety Facts, 2024 Data: Bicyclists; SFMTA Vision Zero High Injury Network (cyclist corridors)

Google's bike route rides past the Tenderloin

…and straight through Civic Center. Same trip: Grace Cathedral → Patricia's Green, Hayes Valley

Google Maps · bicycling · 14 min, 1.9 mi



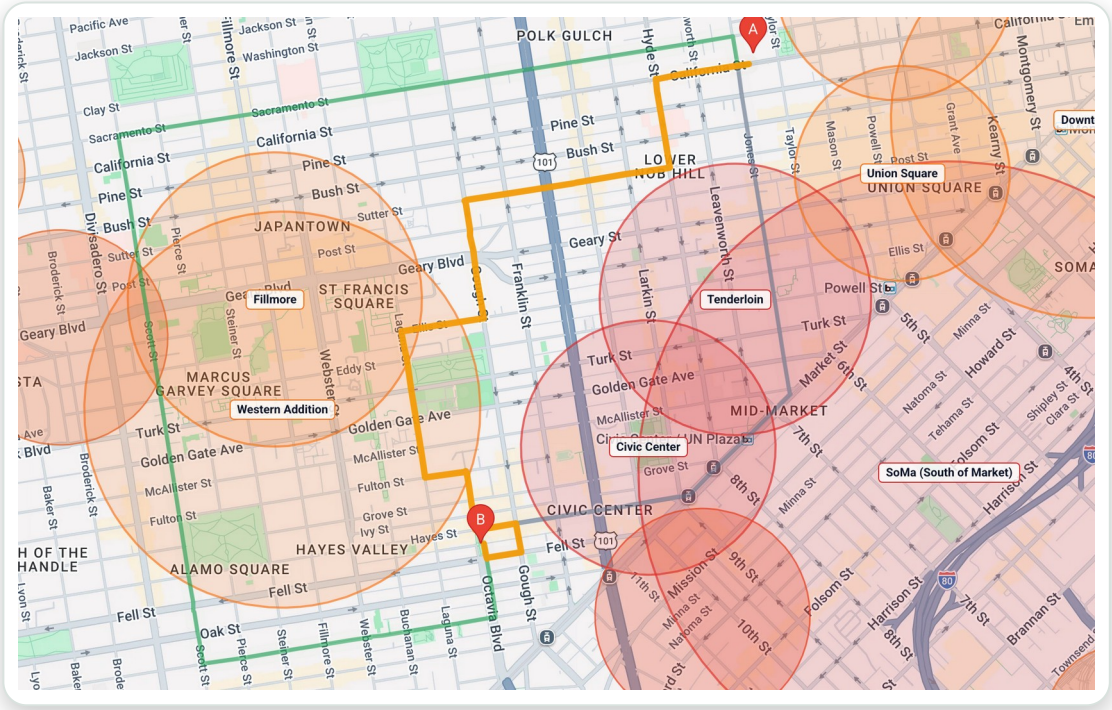
39%

of the ride inside Severe areas: Civic Center 0.5 mi, Tenderloin 0.2 mi

67%

with no protected bike lane (only parts of Polk St are protected)

BetterRoutes · Safest · 22 min, 2.1 mi



0%

inside Severe areas, and it avoids every crash hotspot

−25%

average street danger vs our fastest route, for +5 min

Our own bike router, built on SF's real data

3,558
real bike-injury crashes (DataSF, 2019+)

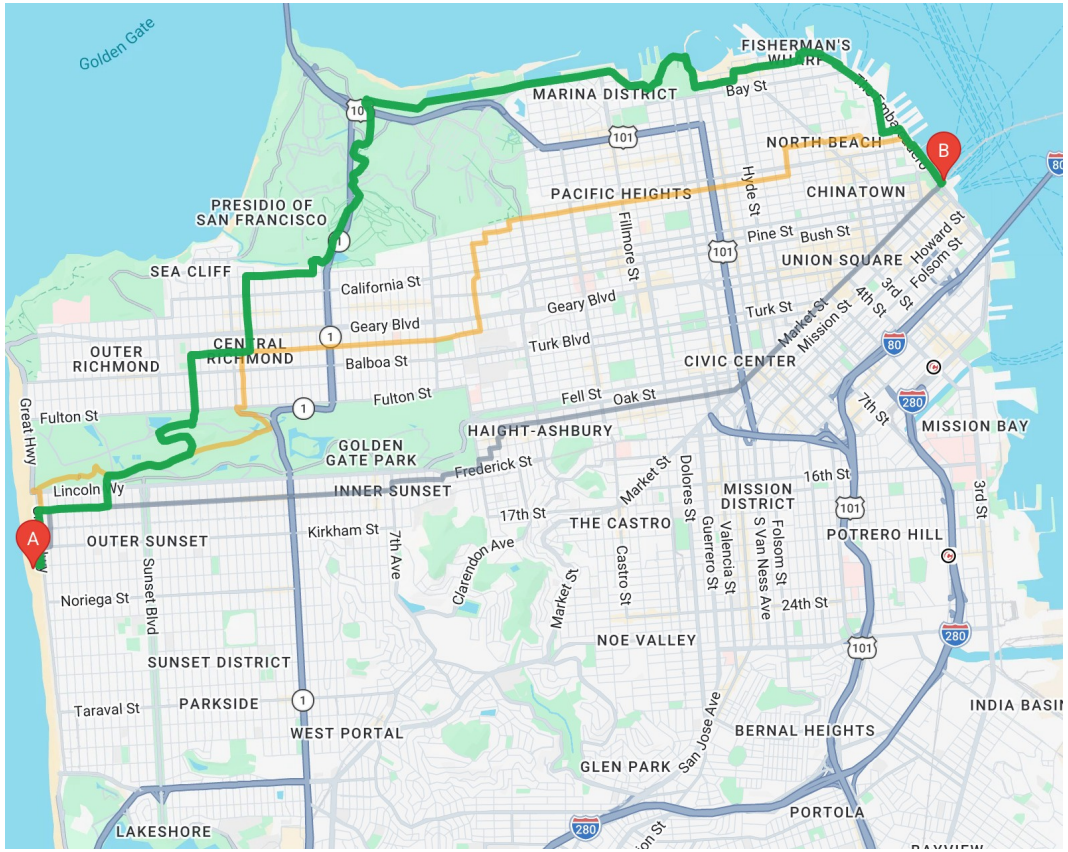
5,455
SFMTA-surveyed bike-lane segments

5,566
highways and fast arterials scored

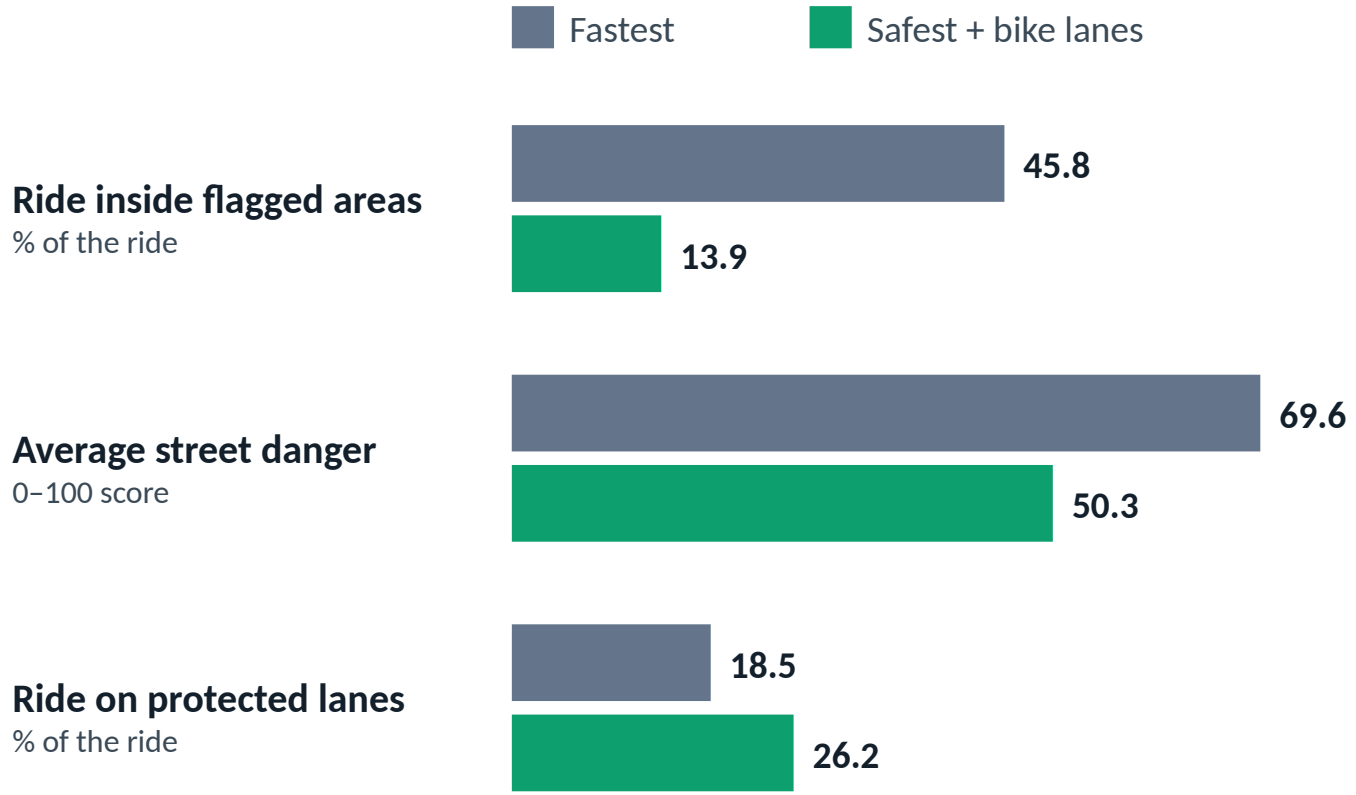
1,314
traffic signals that stop a rider

Every corner
terrain elevation, so hills are real

177,023
LiDAR-measured buildings for 3D



Safer routes cut time in flagged areas by 70%



−70%
time spent inside flagged areas

7 → 1
of 10 trips enter a Severe area

−28%
average street danger; 1.4× protected lanes

+33%
ride time — the honest cost, shown on every route

10 trips typed into the app (first address suggestion), averaged per trip, Oct 14, 2026. Full list in the appendix.

It says what it avoided — and what it cost

Safest + bike lanes

✓ Avoided every Severe area

✓ Avoided every High and Elevated area

✕ Went through 4 crash hotspots (1.3 mi) - no route stays out of all of them

✓ Kept to protected bike lanes where possible

✕ Did not avoid hills

✕ Did not avoid traffic lights

Distance	11.0 mi
Est. time	~96 min
vs fastest route	+26 min
Danger score (avg)	31.9
Worst street on route	76.6
On protected lanes	53%

Rules a rider can read

- Never enters a Severe area
- Avoids High and Elevated areas unless that makes the ride 40%+ longer — then says where and why
- Only claims it "avoided" an area if the fastest route actually goes through it

Avoid hills — Ocean Beach → Ferry Building

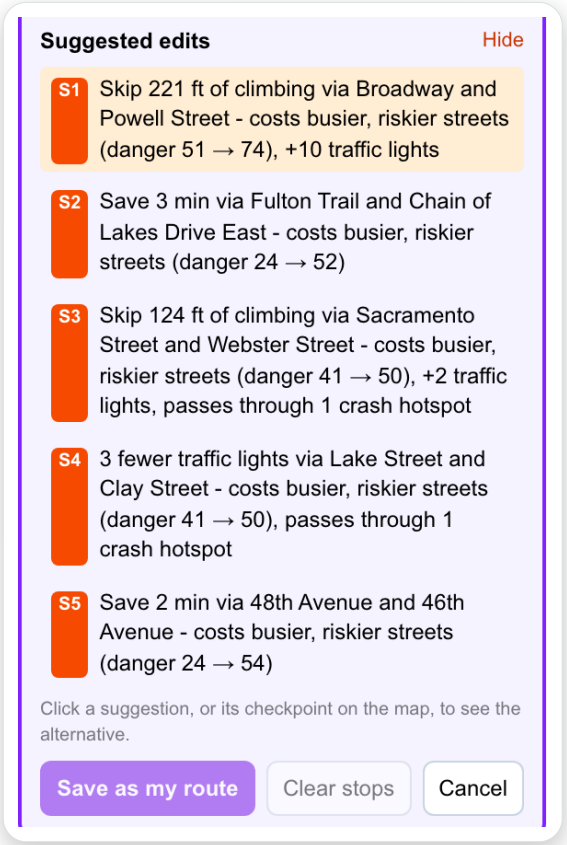
21.8% → 8.6%
steepest climb

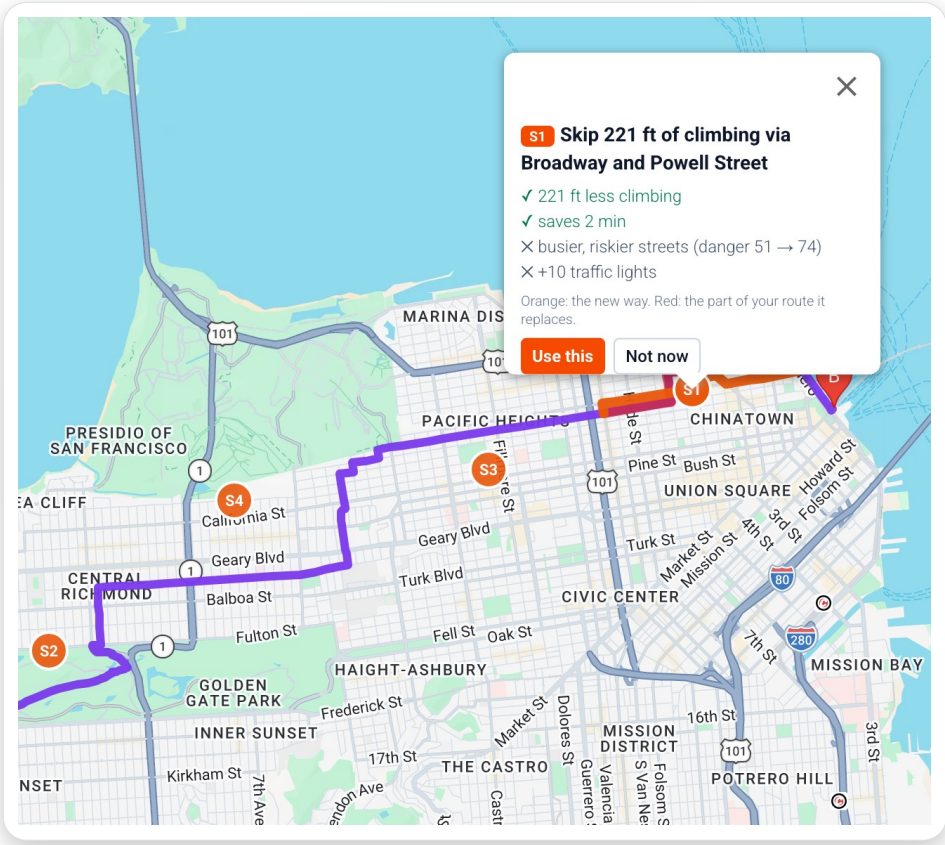
1,027 → 518 ft
total climbing

7 min
quicker, too

▸ 20s demo: Route checklist, then Avoid hills

Make it your route, with every trade-off spelled out





Add, drag or remove stops
It re-plans through them with the same safety rules

Suggested edits
"Skip 221 ft of climbing via Broadway: costs busier streets, +10 lights"

Fewer traffic lights
Fewer stops, and it never overrides safety

Compare side by side
Every stat, one hill scale, best value starred

▸ 20s demo: Edit route + suggested edits

More than safety: real SF data, shaped by you

Avoid hills
Data: USGS elevation at 112,492 points on the street map

21.8% → 8.6%

steepest climb; climbing 1,027 → 518 ft, and 7 min quicker
Try it: Ocean Beach → Ferry Building, Safest

Fewer traffic lights
Data: 1,314 SFMTA traffic signals

35 → 12 lights

Fastest 35, Safest 17, Safest + Fewer lights 12 (2 min quicker than Safest)
Try it: Ocean Beach → Ferry Building

Honest ride time
Data: Every estimate counts each climb and ~12 s per red light

Fastest ≤ Safest

Because times include hills and lights, "Fastest" is never slower than a safer option — checked by automated tests
Try it: Any trip: compare the Est. time on each route

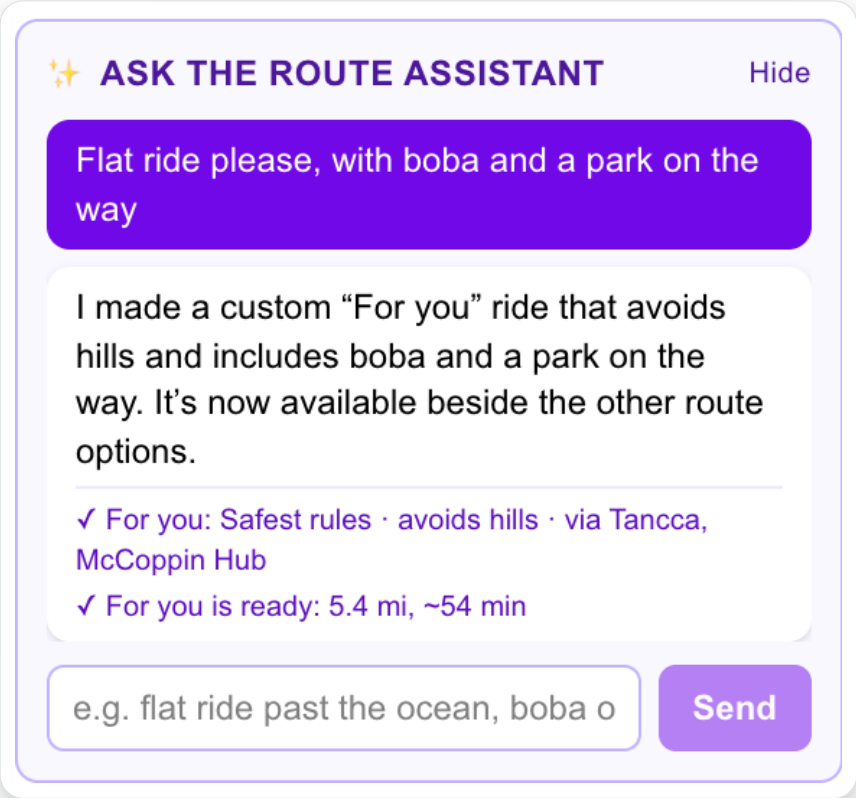
Customize with AI
Data: Same controls you use, plus 3,712 real places

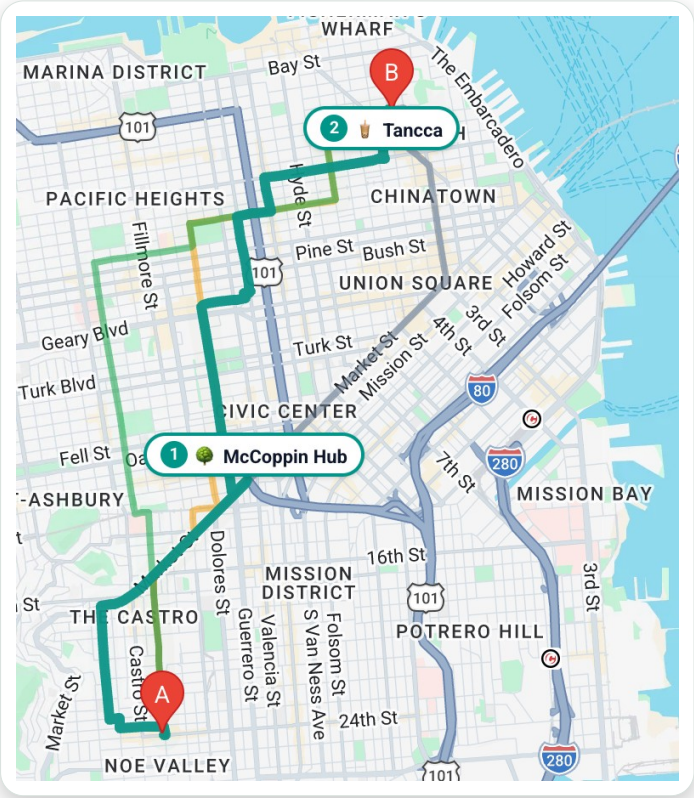
1 sentence

"Flat ride please, with boba and a park on the way" → a For you ride, 5.4 mi, 2 stops
Try it: Noe Valley → North Beach

Danger scores come from our trained model; hills, lights and ride time come straight from measured city data.

Just say what kind of ride you want





Same controls as you

The AI can't break a safety rule — it can only flip the switches a rider can

Visible and undoable

Every change it makes shows up in the sidebar

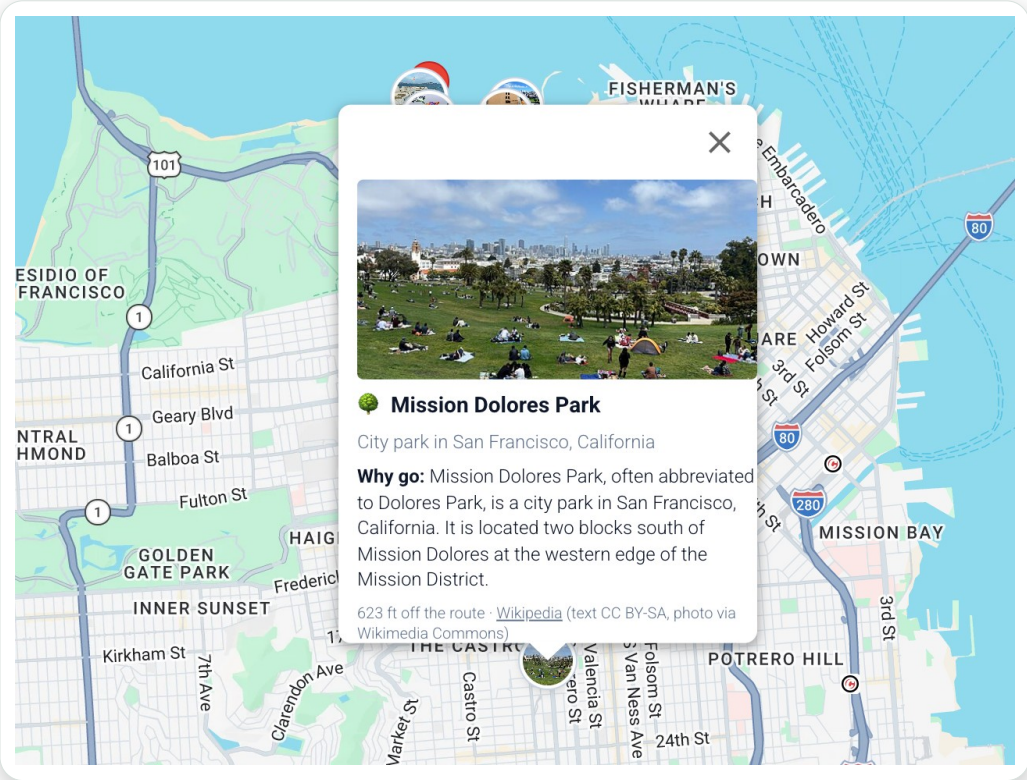
Real places

3,712 spots from OpenStreetMap — never one inside a Severe area

One sentence → a "For you" ride: Safest rules, no big hills, a boba stop and a park, 5.4 mi

▸ 20s demo: Ask the route assistant

Rides around places worth seeing





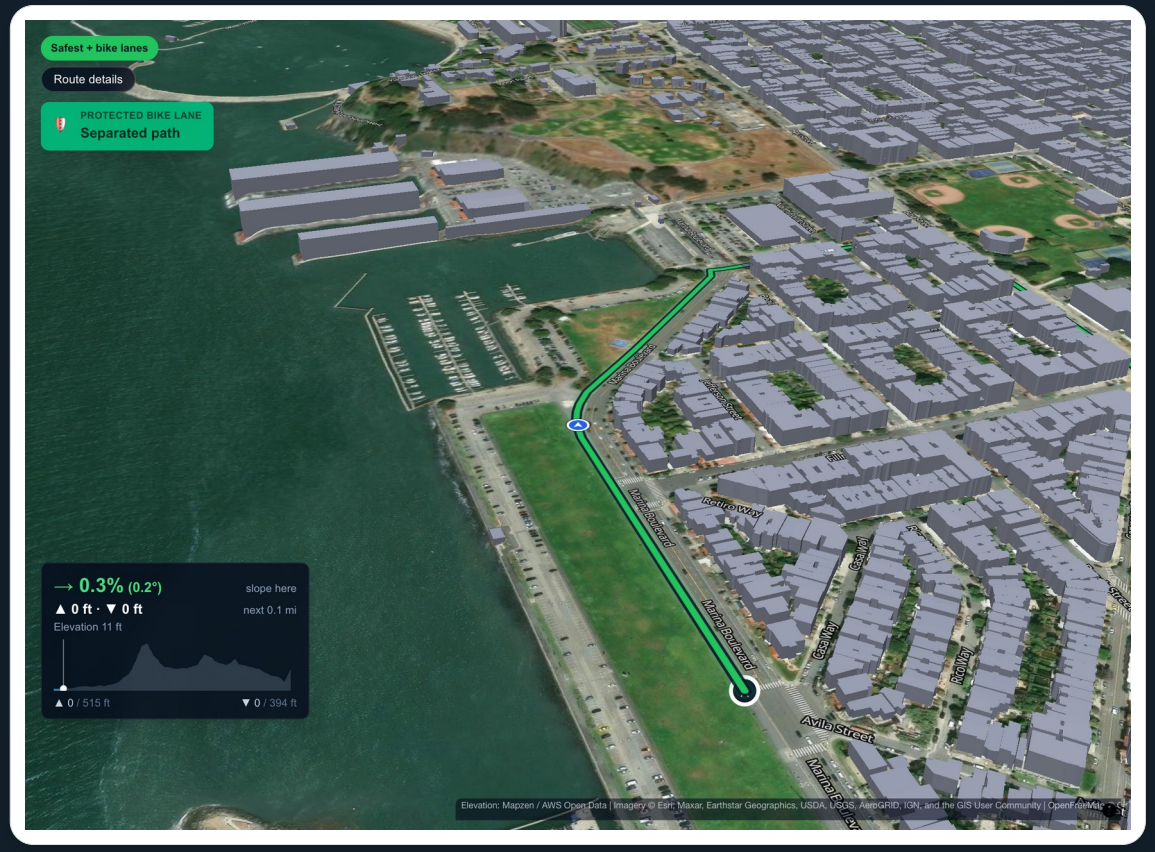
A photo and a reason to stop
Picture, what it is, and why go

Pick what you like
Beaches, boba, coffee, parks, views: a For you ride

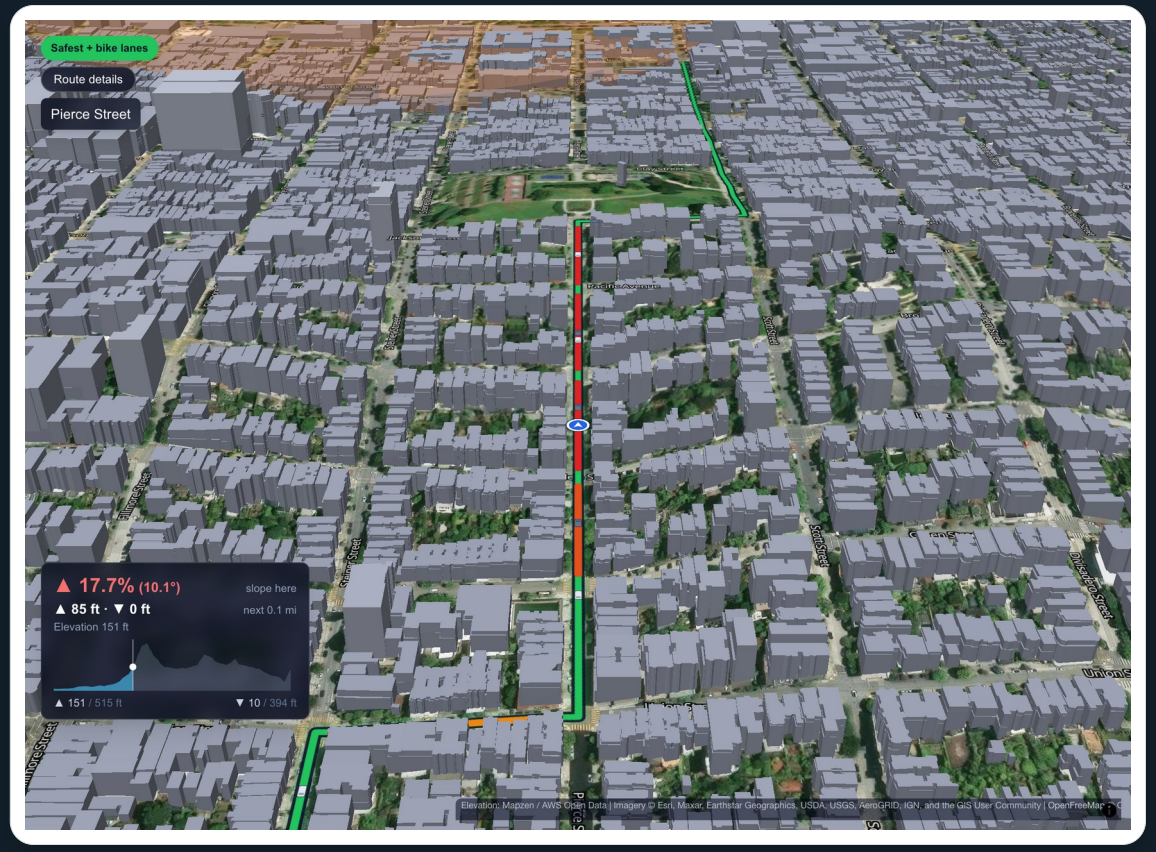
Pinned in the 3D tour
A card pops up as the tour reaches each one

▸ 20s demo: Interests + place photos

Preview the ride before you ride it



Calls out protected lanes the moment you reach them

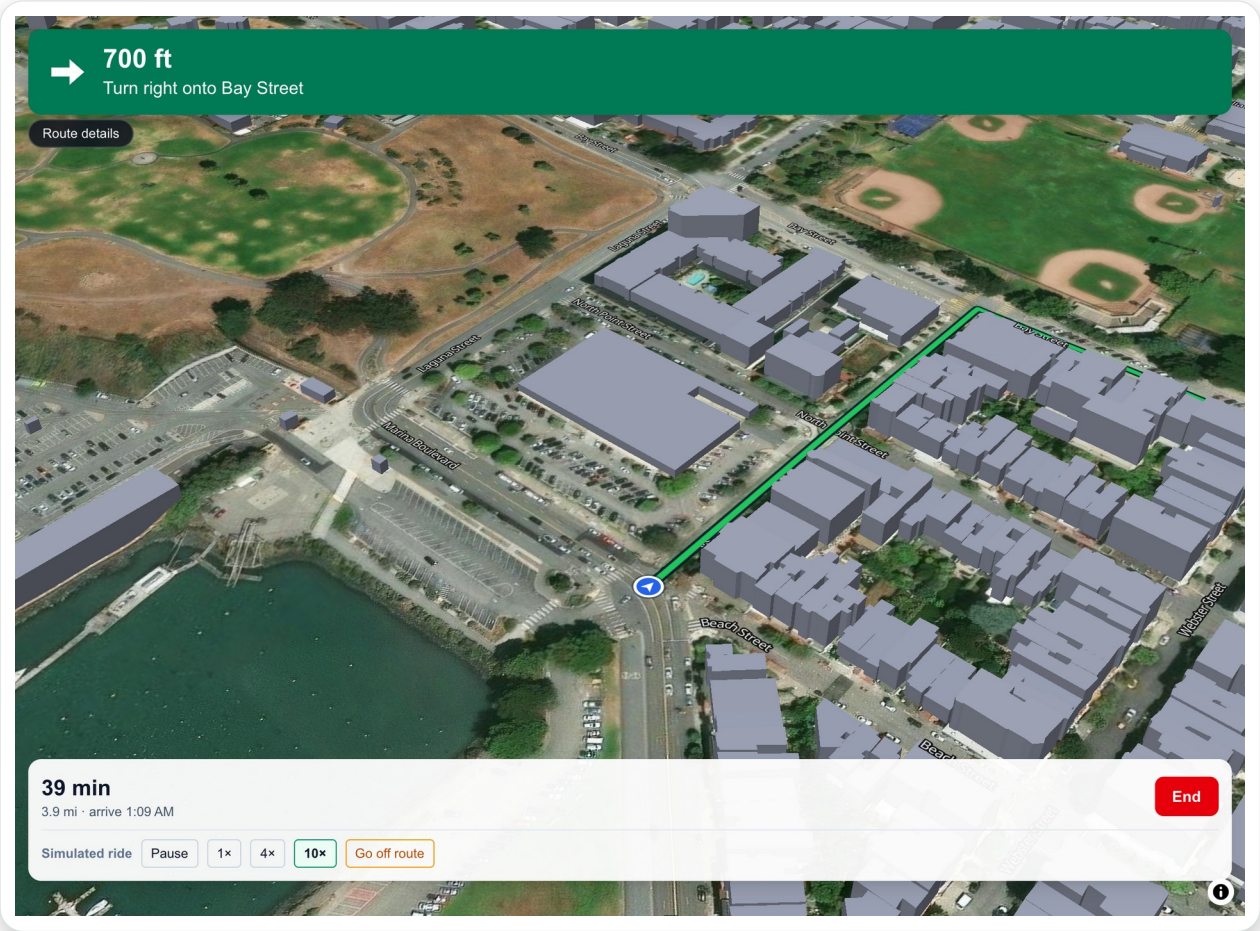


Shows the slope ahead — this block is a 17.7% climb

Real aerial photos · real terrain · San Francisco's own LiDAR building heights

▸ 20s demo: 3D tour with Places on

From plan to the street



Turn-by-turn with live GPS
Works in a phone browser today

Reroutes keep your choice
Go off route and it still plans the safe way back

Compare side by side
Every stat, one shared hill scale, best value starred

Save rides, add photos
Routes, stops and ride photos kept together

▸ 20s demo: Confirm, simulate, reroute, save

Built like a product — and ready for the next city

411 automated tests, all passing

~0.1 s to plan three routes

6 open datasets, no paid data

What's next



Crash rates, not counts
Adjust for how many people ride each street, and learn area risk instead of hand-setting it



Near-miss reports
Add rider reports (BikeMaps.org) to the crash data



More cities
Any city with open crash and bike-lane data — Vision Zero cities first

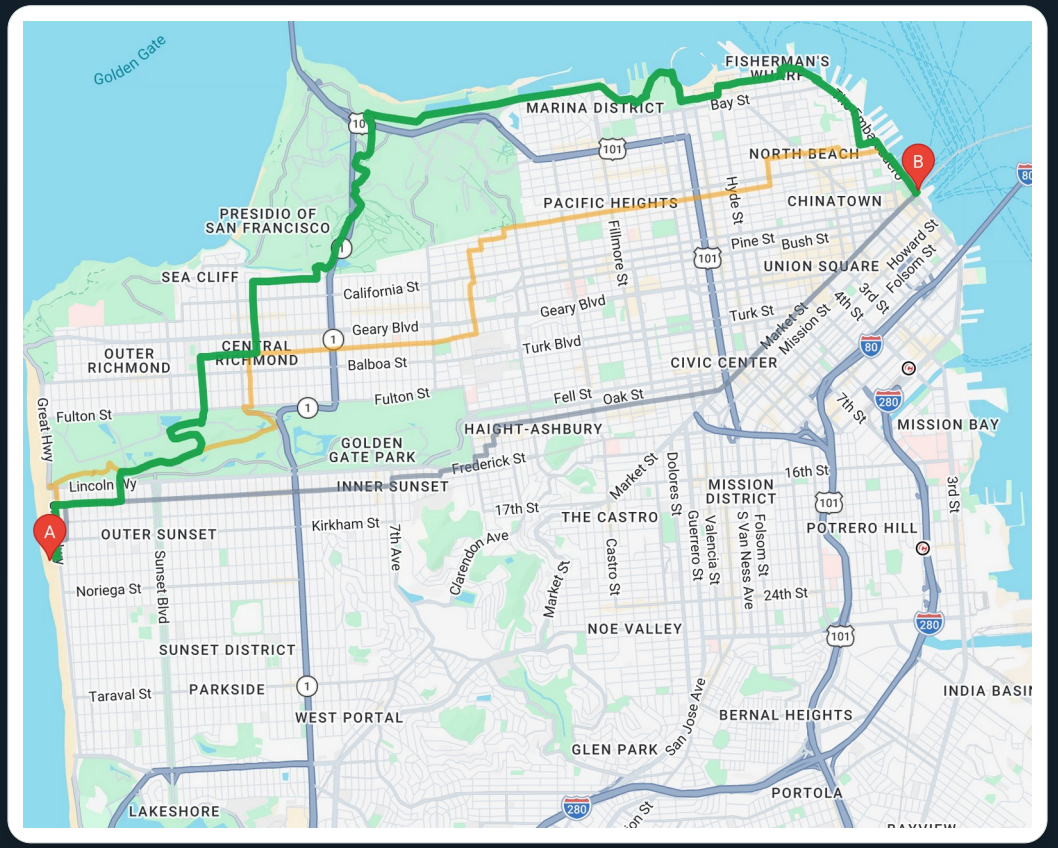


Native app
Background GPS and alerts as you approach a danger spot

▸ 20s demo: 411 tests passing

Safety shouldn't be a setting you have to find

BetterRoutes — the safe route, by default, with the trade-offs in plain sight



Appendix: where every number comes from

Number in the deck	Source	Show it live (From → To, as typed)
1,103 cyclists killed, 52,887 injured (US, 2024)	NHTSA Traffic Safety Facts, 2024 Data: Bicyclists	—
4% of SF street miles hold 55% of severe/fatal bike injuries	SFMTA Vision Zero High Injury Network (cyclist corridors)	Neighborhood view on any trip
About half of big-city adults want to bike but worry about safety	Dill & McNeil, Transportation Research Record 2587 (2016)	—
Google route: 39% in Severe areas, 67% unprotected	Google Maps bicycling directions, captured Oct 14, 2026; SFMTA bike-network data	Grace Cathedral → Patricia's Green
−70% flagged, −28% danger, 1.4× lanes, +33% time, 7→1 of 10	BetterRoutes, 10 trips (next slide)	Any trip on the next slide
Danger 62 → 32, protected 11% → 53%, +26 min	BetterRoutes, Safest + bike lanes vs Fastest	Ocean Beach → Ferry Building
Steepest 21.8% → 8.6%, climbing 1,027 → 518 ft, −7 min	BetterRoutes, Safest with Avoid hills off/on	Ocean Beach → Ferry Building
Traffic lights 35 → 17 → 12	BetterRoutes: Fastest / Safest / Safest + Fewer traffic lights	Ocean Beach → Ferry Building
3,558 crashes · 5,455 lane segments · 5,566 highways · 1,314 signals	DataSF crashes 2019+; SFMTA bike network & signals; OpenStreetMap	lib/data/*.json in the repo
211,221 street segments · 3,712 places · 177,023 buildings	OpenStreetMap; DataSF building footprints (LiDAR)	3D tour shows the building count
411 automated tests, all passing	Vitest suite in the repo	npm test

Appendix: the 10 trips behind the results

From → To (as typed)	Fastest: in flagged areas	Safest + lanes: in flagged areas	Danger	Protected lanes	Time
Ocean Beach → Ferry Building	23.9%	0%	62.1 → 31.9	10.6% → 53.1%	70 → 96 min
Pacific Heights → Mission Street & 17th Street	53.7%	54%	74.6 → 56.8	1.5% → 27.2%	25 → 42 min
Marina Green → Dolores Park	27.2%	17.1%	56.6 → 45.3	0.9% → 18.3%	37 → 43 min
Noe Valley → North Beach	45.2%	14.5%	71 → 48.5	32.9% → 1.3%	43 → 53 min
Grace Cathedral → Hayes Valley	78.3%	0%	88.6 → 69.9	11.8% → 8.4%	15 → 17 min
Chinatown → Dolores Park	58.1%	14.6%	74.1 → 67	39.2% → 19.7%	34 → 41 min
Union Square → Lower Haight	57.2%	13.5%	74 → 68	30.2% → 1.7%	21 → 30 min
Clement Street & 6th Avenue → Potrero Hill	36.8%	12.7%	66.6 → 49.4	4.2% → 12.5%	53 → 66 min
Marina Green → Potrero Hill	47.3%	12.7%	63.6 → 34.7	33.7% → 65%	55 → 72 min
Golden Gate Park → Ferry Building	29.8%	0%	65.3 → 31.9	20.2% → 54.9%	56 → 84 min

Pick the first address suggestion. "Safest + lanes" is the app's "Safest + bike lanes" option. Measured Oct 14, 2026.